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《信息检索与搜索引擎》

课程报告

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真理在辩论中变得更加清晰！具有大型语言模型的多代理系统揭开假新闻的面纱

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摘要

在当今的数字环境中，假新闻通过社交网络的快速传播带来了巨大的社会挑战。大多数假新闻检测方法要么采用传统的分类模型，存在可解释性低、生成能力有限等问题；要么为大型语言模型（LLM）设计特定的提示，让其直接生成解释和结果，无法充分发挥 LLM 的推理能力。受 "真理越辩越明 "这句话的启发，我们的研究引入了一种名为 TruEDebate（TED）的新型多智能体系统，以提高假新闻检测的可解释性和有效性。TED 采用了一种严格的辩论程序，其灵感来自正式辩论环境。我们方法的核心是两个创新组件：DebateFlow Agents（辩论流代理）和InsightFlow Agents（洞察流代理）。DebateFlow 代理将代理分为两队，一队支持新闻真实性，另一队质疑新闻真实性。这些代理进行开场陈述、交叉质证、反驳和总结陈述，模拟类似于人类话语分析的严格辩论过程，从而对新闻内容进行全面评估。同时，InsightFlow 代理由两个专门的子代理组成：合成代理和分析代理。综合代理对辩论进行总结，并提供总体观点，确保评估的连贯性和全面性。分析代理包括角色感知编码器和辩论图，将角色嵌入和模型整合在一起。

我们在 ARG-EN 和 ARG-CN 两个数据集上进行了大量实验，结果表明 TED 框架在各种指标上都超越了传统方法，更重要的是，它增强了可解释的假新闻的能力。我们在两个数据集（ARG-EN 和 ARG-CN）上进行的大量实验表明，TED 框架在各种指标上都超越了传统方法，更重要的是，它通过揭示逻辑推理和结构化辩论过程，提高了可解释的假新闻检测能力，从而得出准确的结论。我们发布了我们的代码，以支持在负责任的信息系统中使用结构化辩论来改进决策的信息系统[1]。

中央监控系统概念

• 计算方法 → 自然语言处理；• 安全和隐私 → 安全和隐私的人类和社会方面。

关键词

假新闻、辩论、大型语言模型、多代理系统

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1 引言

通过社交网络迅速传播的假新闻已变得无孔不入，对个人、弱势社区和社会产生了重大影响。这种广泛传播给舆论治理带来了巨大挑战 [1]，影响选举 [11]，破坏公共卫生政策 [13]，损害公平 [57-59]，威胁社会稳定 [40]。

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[1]https://github.com/LiuYuHan31/TED_ 假新闻删除

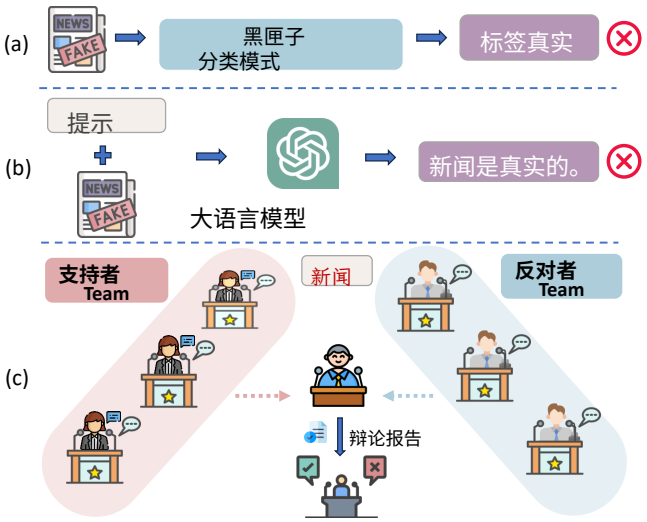


图 1：（a）传统模型主要依靠黑盒分类模型来检测假新闻。(b) LLM 直接输出检测结果，没有充分利用大型模型推理的潜力。(c) 我们的多代理框架模拟正式辩论场景，参与新闻话题讨论，通过辩论推理和可解释性得出检测结果。

传统的假新闻检测方法主要依赖于机器学习模型，尤其是那些基于 BERT [7], 的模型，它们擅长文本分析 [16, 19, 31]。虽然这些模型取得了不俗的成绩，但它们往往存在可解释性低和泛化能力有限的问题，妨碍了它们在不同环境下的有效性。最近的方法试图通过开发特定的提示来生成解释和检测结果，从而利用 LLM [15, 29]。然而，这些方法可能无法充分利用 LLMs 固有的推理能力和复杂的认知过程，经常导致结论过于简化和不成熟。

在寻求真理的过程中，谚语 "真理在辩论中变得更加清晰 "强调了从多角度审视和质疑信息以获得更准确理解的重要性 [2, 21]。在有条理的辩论中交流不同观点往往能完善和澄清观点。然而，现有的假新闻检测模型未能充分结合这一方法，往往将其分析局限于单一的、不受质疑的观点。这种限制妨碍了对新闻内容的全面评估，并可能忽略对准确检测至关重要的细节信号。

为了应对这些挑战，我们引入了一种名为 TruEDebate（TED）的新型多智能体系统。TED 旨在通过采用正式辩论的结构化方法来提高假新闻检测的可解释性和有效性。受 "真理在辩论中更清晰地显现 "这一原则的启发，我们的方法模拟了多个代理之间的动态结构化辩论，每个代理都从不同角度审视新闻。这种辩论结构遵循林肯-道格拉斯辩论（Lincoln-Douglas Debate [8], 的形式，代理参与一系列阶段，提出、质疑和完善他们的论点。这一框架反映了人类通过 "反驳 "来评估信息的认知过程，促进了深度智力参与和对新闻内容的全面评估。在这一框架中，多个代理（每个代理都被分配了不同的角色）参与结构化的辩论过程，以评估新闻条目的真实性。这一辩论过程加强了对检测过程的解释，并反映了人类的辩论认知过程，即对相互冲突的观点进行分析和整合。

TED 框架包括两个创新组件：DebateFlow 代理和 InsightFlow 代理。DebateFlow 代理将多个由 LLM 驱动的代理组织成两个对立团队，一个支持新闻真相，另一个反对新闻真相。这些代理依次进行开场陈述、交叉质证反驳和总结陈词，有效地模拟了严格的辩论环境。这种设置鼓励对不同观点的探索、完善和澄清，并解决了现有模型从单一角度分析信息的局限性。同时，InsightFlow 代理包括两个专门的子代理：综合代理和分析代理。综合代理负责总结辩论内容，提供全面的观点，确保得出连贯、透彻的结论。分析代理由角色感知编码器和辩论图组成，通过互动关注机制整合角色嵌入，并模拟不同辩论角色和论点之间的互动。通过捕捉辩论中的详细关系和内在话语结构，分析代理能有效识别假新闻检测所必需的关键指标。

我们在 ARG- EN 和 ARG-CN 两个数据集上进行了综合实验，以评估 TED 的有效性。结果表明，我们的框架在多个评价指标上都优于传统方法。此外，TED 还促进了结构化辩论程序，增强了逻辑推理能力，有助于得出准确的结论。通过系统化的辩论和深入分析，我们的方法推动了信息系统的发展，使决策过程更加精确和透明，从而支持创建负责任的信息环境。

我们的贡献可以概括为以下几个方面：

- 我们介绍了基于 LLM 的新型可解释假新闻检测框架 TED。TED 不仅能通过结构化辩论进行检测，还能提供可解释的理由，帮助理解检测结果。

{林肯-道格拉斯辩论理论• 认为 "真理在辩论中变得更加清晰"，受该理论的启发，我们在框架中设计了两个关键组件：合成代理和分析代理。综合代理总结辩论内容，综合全面的观点，确保得出连贯的结论。分析代理集成了角色感知编码器和辩论图，整合了角色嵌入，并通过关注机制模拟不同辩论角色和论点之间的相互影响。这种结构使分析代理能够识别对准确识别假新闻至关重要的关键指标。

•在ARG-EN和ARG-CN数据集上进行的大量实验表明，TED的性能优于传统方法，显示了其做出准确透明决策的能力。此外，TED 还能在不同的 LLM 主干网中保持稳定的性能，展示了它在假新闻检测方面的适应性和通用性。

2 相关工作

2.1 假新闻检测

假新闻检测是打击误报的重要一步 [12]。该领域的早期研究包括 Qian 等人的研究 [36], ，该研究侧重于假新闻的早期检测，只涉及检测时可用的新闻文章文本。Yu等人 [49] 采用卷积方法进行错误信息识别。随着该领域的发展，人们引入了各种方法来提高检测效率。例如，Jin 等人 [17] 通过更好地反映人类思维的逻辑过程和实现对微妙线索的建模，向细粒度推理假新闻检测迈进。此外，Zheng 等人 [60] 考虑到多模态数据相互关系的复杂性，研究了如何整合这些数据。最近，随着大型语言模型的兴起，Hu 等人 [15] 将大型和小型语言模型结合起来，仅使用文本内容进行假新闻检测就取得了很好的改进。Li 等人 [25] 设计的基于大型语言模型的代理工作流利用工具学习和知识检索来检测假新闻。Liu等人 [27, 28] 采用多代理系统模拟假新闻的传播和演变。

然而，他们并没有充分利用 LLMs 的潜力。我们的研究通过进一步利用由多个具有 LLMs 的代理组成的辩论系统，增强了假新闻检测的能力。在我们的方法中，我们不需要调用任何外部工具，也不需要基于知识的检索，因此很容易移植到任何假新闻检测场景中。

2.2 有LLM 的多代理讨论

尽管LLM取得了长足进步，但关于多个LLM互动协作解决问题的多代理讨论的研究仍处于早期阶段，其目标是利用超越单个模型的集体能力。ChatEval [3], 是这一领域的一个重要贡献，它设计了一个多代理裁判团队，以独立审议和评估各种模型对开放式问题和标准自然语言生成任务所生成的回答的质量。其他研究则强调代理人角色的多样化，以丰富辩论过程 [24, 33]。Liang 等人 [26] 进一步探讨了角色分配如何影响多代理互动的效率。Chen等人 [4] 提出了ReConcile框架，该框架使用不同的LLM和加权投票系统来达成共识，从而捕捉到更广泛的观点。Du 等人 [9] 利用多代理辩论来提高 LLM 的数学能力。此外，Zhang 等人 [52]从社会心理学角度分析了多代理讨论，以了解社会因素如何影响代理之间的协作和推理过程。

然而，这些作品并没有在其框架内真正描绘辩论过程，也无法从多个角度分析问题。在这些基础上，我们的工作引入了一个结构化辩论框架，利用多代理交互来提高假新闻检测的可解释性和有效性。

2.3 基于 LLM 的多代理系统

凭借 LLMs 强大的文本生成能力，将 LLMs 集成到社会动态模拟中是一个新兴的研究领域，并取得了可喜的成果 [18, 23, 39, 45]。这些基于 LLM 的生成代理在数字环境中表现出色，能够熟练完成自然语言任务 [5, 6, 50] 和推理任务 [51]。Törnberg等人 [42]利用LLM和基于代理的建模来模拟社交媒体，重点研究新闻源算法，并提出了对现实世界的见解。此外，Park 等人 [34] 证明了基于 LLM 的代理可以生成与人类生成的内容不同的社交媒体内容。此外，Qian 等人 [35]设计了一个模拟人类开发团队的虚拟开发公司 ChatDev。

这些利用 LLM 多代理系统解决复杂任务的框架都有一个共同的重要组成部分：多个代理 [3, 14, 30, 41, 46, 54-56] 之间的交互。然而，这些著作只考虑了不同代理之间的信息传递。虽然有些著作声称可以进行多代理辩论，但它们并没有充分描述交互过程中的辩论过程。我们的工作建立了一个标准的辩论场景，不同角色从不同角度对假新闻进行辩论，以确定新闻主题的真实性。据我们所知，这是首次在假新闻背景下模拟社会辩论。

3 方法

3.1 制定任务

我们将假新闻检测重新表述为多个 LLM 代理之间的互动辩论。我们引入了一个由 NLLM 代理组成的系统，表示为 A = （a_1，……，a_N），每个代理在辩论过程中都被分配了独特的角色和立场。给定一个新闻条目 F，所有代理进行结构化辩论，以确定其真实性。代理被组织成两个对立的团队：

• 支持者：认为 F 为真的代理人。• 反对者：认为 F 为假的代理人。

每个代理都在辩论中扮演特定的角色，并参与预定义的阶段。所有互动都记录在辩论日志 D中。在辩论结束阶段，有两类代理对结果进行评估。合成代理对辩论日志 D 进行分析，以生成一份全面的总结报告 S。同时，分析代理利用总结报告 S 和原始新闻 F 来预测真相值。

我们的目标是在这一辩论框架内利用 LLM 的推理能力，提高假新闻检测的有效性和可解释性。

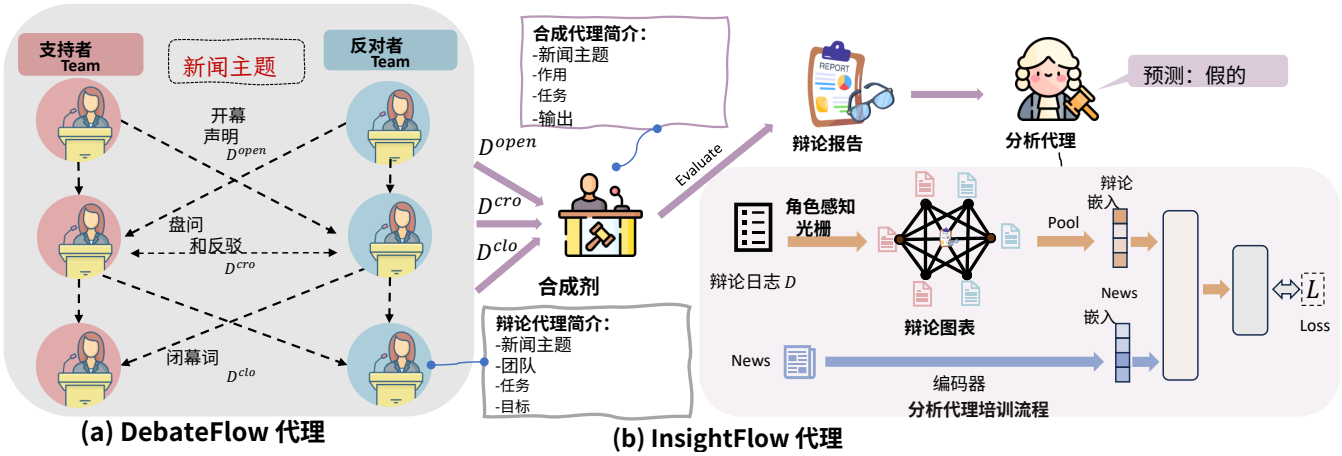


图 2：我们的 TED 框架由两个主要部分组成：(a) DebateFlow 代理和 (b) InsightFlow 代理。DebateFlow 代理模拟辩论场景中的推理过程，包括开篇、反诘问、反驳和结案陈词。InsightFlow 代理由合成代理和分析代理组成。前者根据所有辩论记录生成辩论报告，然后传递给后者以预测新内容是否属实。

3.2 我们的多代理人辩论框架

我们介绍的 TruEDebate（TED）是一个通过模拟多个 LLM 驱动的代理之间的结构化辩论来加强假新闻检测的框架。TED 由两个主要部分组成：

•辩论流程代理：组织成两个对立的团队--正方，负责论证新闻的真实性；反方，负责论证新闻的虚假性。受林肯-道格拉斯（Lincoln-Douglas）辩论理论的启发 [8]，，这些代理参与多阶段的辩论过程，包括开场、盘问、反驳和总结陈词。这种设置反映了正式的辩论环境，鼓励从多个角度对新闻进行彻底审查。

• InsightFlow 代理：由综合代理和分析代理组成。合成代理对辩论进行总结，提供关键论点和反驳论点的连贯概述。分析代理采用角色感知编码器和辩论图来模拟辩论中的互动和话语结构，最终预测新闻的真实性。

如图 2 所示，通过这些代理的互动，TED 促进了深度智力参与和逻辑推理，从而实现了更准确、更可解释的假新闻检测。结构化的辩论过程不仅完善了代理对新闻事件的理解，还为最终决定提供了透明的理由，体现了 "真理越辩越明 "的原则。

3.3 DebateFlow 代理

DebateFlow代理构成了TED框架的核心，负责组织结构化辩论，以评估新闻 F的真实性。代理被分为两个对立团队：正方（Tpro）和反方（Topp），正方认为 F是真的，反方则认为 F是假的。每个团队都由代理人组成，他们被分配了与不同辩论阶段相对应的特定角色。

受林肯-杜格拉斯辩论 [8], 的启发，辩论分为三个主要阶段：开篇、盘问和反驳以及结尾。让 $\Phi = \{\phi_1, \phi_2, \phi_3\}$ 表示这些阶段。在每个阶段 $\phi_k \in \Phi$ 中，两队的代理人根据各自的角色参与。在开场阶段（ϕ_1），代理人提出支持本队立场的初始论据。每个代理人 a_i 根据新闻 F和本队立场生成一个论据 $d(1)_i$：

$$d_i^{(1)} = f_{arg}(F, Stance_i), \tag{1}$$

其中，f_{arg} 代表由 LLM 执行的论证提示功能，$Stance_i \in \{True, Fake\}$对应于团队的立场。

在交叉询问和反驳阶段（ϕ_2），代理人通过提问和反驳来质疑对方的论点。一队的代理人a_j 针对对方在前一阶段提出的论点提出问题或反驳$d(2)_j$：

$$d_j^{(2)} = f_{reb}(D^{(1)}, Stance_j), \tag{2}$$

其中，f_{reb} 是反诘问和反驳提示函数，$D(1)$ 是阶段 ϕ_1 的参数集。

在结束阶段（ϕ_3），代理人总结自己团队的立场并强化关键点。每个代理人 a_k 都要做总结发言 $d(3)_k$：

$$d_k^{(3)} = f_{clo}(D^{(1)} \cup D^{(2)}, Stance_k), \tag{3}$$

其中，f_{clo}是收尾提示函数，$D(1) \cup D(2)$代表前几个阶段累积的辩论内容。

辩论日志 D 累积了辩论期间的所有互动：

$$D = \bigcup_{k=1}^{3} D^{(k)}, \quad \text{with} \quad D^{(k)} = \{d_i^{(k)} \mid a_i \in \mathcal{A}^{(k)}\}, \tag{4}$$

1: 输入新闻条目 F2： 输出：预测新闻 F的真相值$\hat{y}$，解释性理由 R3： 初始化 DebateFlow 代理：4: 将代理分成两队：

- 组成部分：支持 F 为真
- 反对者论证 F 是假的

5：在每个小组内分配角色（例如，开场发言人、提问者、结辩发言人） 6：辩论流程代理：7：对于每个辩论阶段 ϕ，在 [开篇陈词、交叉质询、反驳和结辩]中进行 8：代理 i 根据角色提出论点或反驳 9：在辩论日志 D_ϕ中记录互动 10：结束 11：InsightFlow 代理：12：合成代理： 13：分析辩论日志 $D = \{D_\phi\}$，生成评估报告 S14：分析代理： 15：对于 D 中的每个辩论互动（节点） n_i，执行 16：使用角色感知编码器获取节点 n_i 嵌入 h_i 17：结束 for 18：使用节点 $V = \{h_i$构建 Dabate 图 $G = （V, E）$ $G = (V, E)$19：对 G 应用 GAT，获得图嵌入 g20：对新闻条目 F 编码，获得内容嵌入 e_F21：组合表示 $c \leftarrow \text{InteractiveAttention}(g, e_F)$ 22：使用 c预测真值$\hat{y}$ 23：返回预测真值$\hat{y}$，解释原因$R = \{S, D\}$

算法 1：TruEDebate (TED)：多代理辩论框架--假新闻检测工作

其中 A (k) 是参与阶段 ϕ_k 的代理人集合。

每个代理人的贡献 d（k）$_i$取决于新闻项目 F、其团队的立场、其在辩论中的角色以及之前的辩论背景$D^{<k}$：

$$d_i^{(k)} = f_{\text{LLM}}(F, D^{<k}, \text{Role}_i, \text{Stance}_i), \tag{5}$$

其中， $D^{<k} = \bigcup_{m=1}^{k-1} D(m)$ 表示截至阶段 ϕ_{k-1} 的辩论历史，而角色$_i$则指定了代理的角色。

整个算法过程可在算法 1 中描述。这种结构化的互动确保了代理同时考虑新闻内容和不断变化的辩论背景，促进了推理的连贯性和深度。通过参与这一过程，DebateFlow 代理模拟了对新闻项目 F 的严格审查，反映了人类辩论的动态，并促进了从多个角度进行综合评估。

整套功能提示可在我们的代码文件中找到。

3.4 InsightFlow 代理

在辩论流代理进行辩论之后，洞察流代理对积累的话语进行分析，对新闻 F的真实性做出最终预测。该部分由两个主要代理组成：合成代理和分析代理。

3.4.1 综合代理。合成代理处理辩论日志 D（其中包含辩论阶段的所有互动），生成一份简明而全面的摘要 S，总结双方提出的关键论点和反驳意见。该摘要将从假新闻检测的多个角度进行评估，以确保做出全面的判断。形式上，合成代理的核心功能提示定义如下：

> 报告应详细说明您对辩论的评估。通过检查以下内容，重点评估该话题所涉及新闻的真实性：1.新闻是否包含具体细节和可核实信息。2.新闻是否引用了可靠的消息来源或新闻组织。3.新闻的语气和风格，真实的新闻通常更加客观和中立。4.是否使用情绪化语言，这可能是假新闻的特征。5.新闻中的信息是否可以通过其他可靠渠道得到证实。

综合代理的目的是概括辩论的关键要点，提供一个连贯的概述，突出双方论点的优缺点。这个综述 S 连接了辩论话语和后续分析流程，确保最相关的信息随时可供进一步评估。

3.4.2 分析代理。分析代理将辩论互动与原始新闻内容相结合，以预测新闻条目 F的真值$\hat{y}$。该代理结合使用了角色感知编码器、辩论图和新闻-辩论互动关注机制，以捕捉辩论的语义内容和关系动态，以及它们与新闻内容的相关性。

角色感知编码器。为了捕捉不同辩论角色的显著特征，我们设计了一种角色感知编码器。对于每个辩论互动 d_i，节点表示是通过将编码器的 [CLS]令牌嵌入与预测的角色嵌入相结合来构建的：

$$\begin{aligned} \mathbf{h}_i^{\text{enc}} &= \text{Encoder}(\text{Text}_i), \\ \mathbf{r}_i^{\text{proj}} &= \mathbf{W}_{\text{role}}\, \mathbf{e}_i, \\ \mathbf{h}_i^{\text{node}} &= \left[\mathbf{h}_i^{\text{enc}} ; \mathbf{r}_i^{\text{proj}} \right], \end{aligned} \tag{6}$$

其中，henc$_i$ $\in \mathbb{R}^{d_h}$ 是来自编码器的 [CLS]令牌表示，e$_i$ $\in \mathbb{R}^{d_r}$ 是角色 r_i的嵌入，Wrole $\in \mathbb{R}^{d_h \times d_r}$投射角色嵌入以匹配编码器的维度， [;] 表示向量连接。由此产生的节点代表 hnode$_i$ 整合了每次辩论互动的语义内容和特定角色信息。

辩论图。编码后的节点表示被构造成一个辩论图 $G = (V, E)$，其中节点 $V = \{h\text{node}_i\}$ 为 $G = (V, E)$。

代表辩论互动，边 E 代表互动之间的关系，如顺序或明确引用。

我们采用图注意网络（GAT）[43] 来处理节点表征。GAT 层通过关注每个节点的邻居来更新每个节点的表征：

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\right), \tag{7}$$

其中，$\mathrm{h}(l)_i$ 是节点 i 在层 l 中的表示，$\mathrm{N}(i)$ 表示节点 i 的邻居集，$\mathrm{W}(l)$ 是可学习的权重矩阵，σ 是激活函数，$\alpha(l)_{ij}$ 是注意力系数。通过堆叠多个 GAT 层，该模型可以捕捉辩论中的高阶交互和关系动态。

辩论-新闻互动关注。为了将辩论表征与新闻内容结合起来，我们采用了一种互动关注机制。首先，我们对新闻内容 F 进行编码，然后应用全局池化，得到图层面的辩论表征 g：

$$\begin{aligned} \mathbf{e}_F &= \mathrm{Encoder}(F), \\ \mathbf{g} &= \mathrm{GlobalPool}(\mathbf{H}^{(L)}), \end{aligned} \tag{8}$$

其中，$\mathrm{H}(L)$ 是第 L 个 GAT 层的输出，GlobalPool 表示全局均值池操作。

接下来，我们将辩论表征和新闻内容嵌入投影到一个共同的维度上：

$$\begin{aligned} \mathbf{g}^{\mathrm{proj}} &= \mathbf{W}_g\mathbf{g}, \\ \mathbf{e}_F^{\mathrm{proj}} &= \mathbf{W}_e\mathbf{e}_F, \end{aligned} \tag{9}$$

其中 $\mathrm{W}g$ 和 $\mathrm{W}e$ 是投影矩阵。

然后，我们利用多头注意力（MHA）计算出互动表示 c，以捕捉脱口秀和新闻内容之间的互动：

$$\mathbf{c} = \mathrm{MHA}(\mathbf{e}_F^{\mathrm{proj}}, \mathbf{g}^{\mathrm{proj}}, \mathbf{g}^{\mathrm{proj}}), \tag{10}$$

最后，我们将预测辩论表示法与互动表示法合并，得到综合表示法：

$$\mathbf{h} = [\mathbf{g}^{\mathrm{proj}}; \mathbf{c}]. \tag{11}$$

这种互动式关注使模型能够关注与新闻内容相关的辩论的不同方面，有效地整合来自两个来源的信息。

代理训练。组合表征 h 通过全连接层处理，然后通过软最大激活生成类别的预测概率分布。模型的训练方法是尽量减少预测概率与实际标签之间的交叉熵损失：

$$\begin{aligned} \hat{\mathbf{y}} &= \mathrm{softmax}(\mathbf{W}_{\mathrm{fc}}\mathbf{h} + \mathbf{b}_{\mathrm{fc}}), \\ \mathcal{L} &= -\sum_k y_k \log \hat{y}_k, \end{aligned} \tag{12}$$

其中，Wfc 和 bfc 表示分类层的权重和偏置，ˆy 表示预测概率分布，y_k 是类别 k 的基本真实标签，$\hat{y}_k$ 是类别 k 的预测概率。

4 实验

4.1 实验设置

4.1.1 数据集为了进行评估，我们使用了两个数据集：ARG-EN 和 ARG-CN [15]。这两个数据集分别来自中文数据集微博 21 [32] 和英文数据集 GossipCop [38]。在之前的工作 [15], 中，他们通过重复数据删除和时间分割来处理这些数据集。这种方法有助于防止因潜在的数据泄漏而高估 SLM 性能。我们在实验中使用了这两个数据集。表 2 总结了数据集的统计详情。

4.1.2 基线根据之前的工作 [15], , 我们对三种方法进行了实验比较。

- 仅 LLM 方法：直接使用 GPT-3.5-turbo 和 GPT-4o- mini 作为假新闻检测的 LLM。
- 纯小语言模型（SLM）方法：小语言模型（SLM）与基于 BERT 的模型一样，通常在假新闻检测任务中表现良好。在这一类别中，我们包括以下方法：

(1) Bert [7]：使用经过微调的 vanilla BERT-base 模型检测假新闻。

(2) EANN [44]：该模型采用辅助对抗训练法，以发表年份作为辅助任务的标签，有效地分离和减少与事件相关的特征。

(3) Publisher-Emo [53]：该模型整合了一系列情感属性和文本特征，用于检测假新闻。

(4) ENDEFA [61]：该模型利用因果学习来消除实体偏差，从而增强了其在假新闻数据集上的泛化能力。该类别中的所有方法都使用相同的 SLM（BERT）进行文本编码。

- LLM+SLM 方法。单纯的 LLM 通常在任务中表现不佳。因此，许多研究将 LLM 与 SLM 结合起来，以提高性能。

(1) Bert+理由：它结合了新闻和理由编码器的特征，将其输入 MLP 进行预测。

(2) SuperICL [47]：它利用 SLM 作为插件，通过将每个测试样本的预测和置信度纳入提示，增强 LLM 的情境学习能力。

(3) ARG [15]：该模型为假新闻检测设计了一个自适应理由指导网络，其中 SLM 从 LLM 提供的理由中选择性地获得新闻分析的见解。

(4) ARG-D [15]：它是通过蒸馏创建的 ARG 的无理版本，设计用于不需要查询 LLM 的成本敏感型场景。

4.1.3 实现细节。我们使用 Python 来实现我们的框架 TED。使用的 LLM 是通过 OpenAI API 调用访问的 GPT-4o-mini，这是因为它具有代表性且调用方便。我们使用 Python 库 Mesa [20] 定义了代理和它们所处的场景。对于分析代理，我们使用了微调 BERT 模型，因为最近的研究 [19]强调了该模型在这一领域的广泛应用。在实验中，我们使用了 Transformers 软件包中的中文-bert-wwm-ext 模型和 bert-base-uncased 模型进行英文评估。我们使用 Adam [22]，通过全面的网格搜索进行优化，以达到最佳效果。

表 1：对 TED 以及纯 LLM、纯 SLM 和 LLM+SLM 方法的性能进行了评估。基线的所有结果（GPT-4o-mini 和 ChatEval 的结果除外）均来自 Hu 等人的研究 [15]。macF1-分数和准确率的前两项结果均突出显示，每个类别中最高的结果用粗体表示，第二高的结果用下划线表示。粗体数字表示基线模型的改进具有统计学意义（t 检验，p 值<0.01）。

模型		ARG-EN				ARG-CN			
		macF1	Acc.	$F1_{真实}$	$F1_{伪造}$	macF1	Acc.	$F1_{真实}$	$F1_{伪造}$
仅限法学硕士	GPT-3.5-turbo	0.702	0.813	0.884	0.519	0.725	0.734	0.774	0.676
	GPT-4o-mini	0.691	0.845	0.909	0.472	0.725	0.746	0.780	0.670
仅限 SLM	Bert	0.765	0.862	0.916	0.615	0.753	0.754	0.769	0.737
	EANN	0.763	0.864	0.918	0.608	0.754	0.756	0.773	0.736
	出版商-Emo	0.766	0.868	0.920	0.611	0.761	0.763	0.784	0.738
	结束语	0.768	0.865	0.918	0.618	0.765	0.766	0.779	0.751
LLM+SLM	伯特 + 理由	0.777	0.870	0.921	0.633	0.767	0.769	0.787	0.748
	超级ICL	0.736	0.864	0.920	0.551	0.757	0.759	0.779	0.734
	ARG	0.790	0.878	0.926	0.653	0.784	0.786	0.804	0.764
	ARG-D	0.778	0.870	0.921	0.634	0.771	0.772	0.785	0.756
多方代理	聊天评估（逐个评估）	0.733	0.860	0.919	0.546	0.694	0.717	0.778	0.611
	聊天评估（同时聊天）	0.738	0.869	0.923	0.553	0.694	0.719	0.780	0.608
	特德（我们的）	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**

表 2：假新闻检测数据集的统计数据。

	ARG-CN			ARG-EN		
	火车	**Val**	**Test**	**火车**	**Val**	**Test**
Real	2,331	1,172	1,137	2,878	1,030	1,024
Fake	2,873	779	814	1,006	244	234
总计	5,204	1,951	1,951	3,884	1,274	1,258

确定最佳学习率。使用测试结果中的最佳验证检查点来评估性能。对于分析代理，实验使用 PyTorch 进行，其中 HuggingFace Transformers 库用于预训练模型，PyTorch Geometric 用于图神经网络组件。文本数据使用适合数据集语言的 BERT 标记器进行标记。辩论互动使用角色感知嵌入编码，新闻内容使用 BERT 模型单独编码。此外，每个辩论实例都采用了固定的图结构，节点代表辩论互动，边代表这些互动之间的逻辑流和关系。边是根据辩论顺序和角色之间的互动预先定义的。所有实验都在配备英伟达 3090 GPU 的机器上运行，以加速训练。具体的实验实施细节可参见我们的代码。

4.2 主要成果

表 1 列出了我们提出的 TED 框架和其他各种模型（如纯 LLM、纯 SLM 和 LLM+SLM 方法）在 ARG-EN 和 ARG-CN 数据集上的性能。结果表明：(1) TED 优于所有其他方法，在所有数据集上都获得了最高的 macF1 分数和准确率、展示了其在检测假新闻方面的卓越能力。这凸显了我们的多代理辩论法在全面分析和判断新闻真实性方面的功效。(2) 与 TED 相比，仅使用 LLM 的方法（不使用结构化论证的 LLM）表现逊色。(3) 仅使用 SLM 的方法之间的比较显示，每种变体的性能都在逐步提高，但都没有达到 TED 的性能水平。这表明，传统的 SLM 虽然有能力，但与 TED 的多代理系统所提供的强大而全面的分析框架相比，还是有差距的。(4) LLM+SLM 方法试图结合 LLM 与 SLM 的能力，但表现各异。例如，ARG 模型显示出相当的有效性，但仍未达到 TED 的水平。这表明，虽然 LLM 和 SLM 的整合带来了好处，但 TED 的结构化辩论形式对最佳性能至关重要。

此外，为了评估 TED 框架在假新闻检测方面的有效性，我们将其与 ChatEval 框架 [3], 进行了比较，后者使用 "一对一"（One-by-One）和 "同时对话"（Simultaneous-Talk）策略进行多代理辩论。将这些策略应用于假新闻检测任务，如表 1 所示，在所有指标上，TED 始终优于 ChatEval。TED 的正式辩论结构促进了代理之间更多的互动推理，从而可以对新闻内容进行更深入的分析。相比之下，ChatEval 的策略在假新闻检测方面没有达到同样的深度。这些结果凸显了 TED 在提高多代理系统假新闻检测的准确性和可靠性方面的优势。通过融合结构化和动态的交互，TED 不仅提高了检测的准确性，还确保了对新闻项目进行更彻底的检查。

表 3：我们的模型 TED 在 ARG-EN 和 ARG-CN 数据集上的消融研究。粗体数字表示消融模型的改进具有统计学意义（t 检验，p 值 <0.01）。

	ARG-EN				ARG-CN			
	macF1	Acc.	$F1_{真实}$	$F1_{伪造}$	macF1	Acc.	$F1_{真实}$	$F1_{伪造}$
完整模型 (TED)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**
无 *DebateFlow* 代理	0.780	0.874	0.918	0.642	0.767	0.781	0.795	0.739
不含合成剂	0.792	0.881	0.924	0.660	0.771	0.787	0.801	0.741
不含分析剂	0.729	0.856	0.915	0.543	0.737	0.736	0.785	0.689

5 分析与讨论

5.1 消融研究

我们进行了一项消融研究，以评估 TED 框架中每个组成部分的影响。结果表明，去掉 DebateFlow 代理后，macF1 分数在 ARG-EN 和 ARG-CN 上分别降低了约 2.3% 和 2.8%。这一下降凸显了结构化辩论流程在促进多角度综合分析方面的重要性，而多角度综合分析对于准确判断新闻项目的真实性至关重要。去掉合成代理后，macF1 分数略有下降，在 ARG-EN 中下降了 1.1%，在 ARG-CN 中下降了 2.4%。这凸显了合成代理在将辩论互动整合为连贯的摘要方面的关键作用，从而提高了模型有效整合不同论点以进行精确分类的能力。省略分析代理导致的得分下降最为明显，macF1 在 ARG-EN 上的得分下降了 8.0%，在 ARG-CN 上的得分下降了 4.5%，这强调了分析代理在将辩论见解与新闻内容相结合方面的关键作用。总之，消减结果证实了 TED 框架的每个组件对于实现假新闻检测的高性能都是不可或缺的。辩论流代理（DebateFlow Agents）实现了稳健的多代理互动；合成代理（Synthesis Agent）确保了辩论结果的有效整合，而分析代理（Analysis Agent）则将这些见解与新闻内容整合在一起，以获得精确的结果。

5.2 不同骨干网的性能

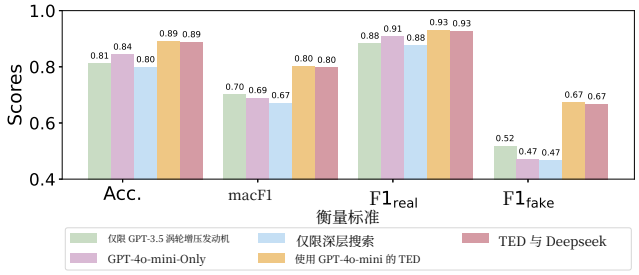


图 3：我们的框架 TED 在使用 GPT-4o- mini 和 Deepseek 作为闭源骨干网时的性能。

5.2.1 闭源模型的性能。为了评估 TED 框架的易用性和可变性，我们在 ARG-EN 数据集上进行了实验，用不同的闭源模型替换了 DebateFlow 代理和合成代理、即 GPT-4o-mini 和 Deepseek[2]。图 3 显示，在所有评估指标上，与基线 LLM-Only 模型相比，无论使用哪种闭源模型，TED 都能持续提升性能。这一结果凸显了该框架对不同大规模模型的适应性，表明 TED 有效地利用了各种闭源骨干网的独特优势。此外，我们还观察到，使用 GPT-4o-mini 的 TED 和使用 Deepseek 的 TED 性能相当，GPT-4-mini 略胜一筹。这表明，虽然这两个模型具有竞争力，但 GPT-4o-mini 似乎在某些方面略胜一筹，尤其是在处理复杂推理任务方面。尽管如此，两个骨干网的结果都强调了 TED 在提高假新闻检测性能方面的稳健性，而与所使用的特定闭源模型无关。这些发现强调了 TED 并不局限于单一的骨干网，它可以有效地提高不同闭源模型的性能。TED 在使用 GPT-4o-mini 时略胜一筹，这进一步证实了 TED 可以利用每个模型推理能力的固有优势这一观点。这种适应性确保了 TED 在实际应用中的有效性和可靠性。

总之，TED 框架被证明既有效又适应性强，在各种闭源模型中都能保持强劲的性能。这些结果验证了 TED 在各种应用场景中的实用性，使其成为一种用途广泛的假新闻检测工具。

表 4：分别以 Qwen2.5 和 Llama3.1 家族模型为骨干的 TED 性能。带下划线的数字表示使用开源模型时的最佳性能，且具有统计学意义（t 检验，p 值<0.01）。

骨架	#Param	macF1	**Acc**	$F1_{\textbf{real}}$	$F1_{\textbf{fake}}$
GPT-4o-mini	-	0.803	0.892	0.932	0.674
Qwen 2.5 [48]	7B	**0.794**	**0.887**	**0.926**	**0.661**
拉马 3.1 [10]	8B	0.787	0.879	0.920	0.653
拉马 3.1 [10]	70B	0.790	0.884	0.923	0.657

5.2.2 开源模型的性能。为了进一步评估 TED 框架的适应性和鲁棒性，我们在 ARG-EN 数据集上使用多个开源语言模型测试了其性能，包括 Qwen-2.5 [48] 和 Llama 3.1 系列 [10]。如表 4 所示，TED 持续改善了 ARG-EN 数据集的性能。

[2]https://www.deepseek.com/

在所有开源模型中，TED 的表现都很突出，这表明无论使用何种特定语言模型，它都能显著增强假新闻检测能力。这表明，TED 不受骨干模型的可用性或类型的限制，因此成为适用于各种实际应用的通用而有效的工具。一般来说，TED 在 Qwen 2.5 中的性能略微优于 Llama 3.1 变体。虽然性能差异很小，但这表明 Qwen 2.5 特别适合 TED 的辩论驱动框架。它可以更好地处理复杂的论证和上下文，这对于 TED 所依赖的结构化评估至关重要。另一方面，使用 Llama 3.1 模型的 TED 也表现出色，所有配置的结果都优于基线方法，进一步验证了 TED 在不同模型中的有效性。虽然开源模型的性能低于 GPT-4o-mini 的结果，但这可能是由于这些模型的底层推理能力不同造成的。开源模型虽然能力很强，但在细微推理方面可能不如 GPT-4o-mini 等封闭源模型。尽管如此，即使使用这些开源骨干，TED 仍能继续产生强大的结果，这表明它仍然可以利用每个模型的优势来实现高水平的准确性和可靠检测。

总体而言，结果凸显了 TED 的稳健性和多功能性。它不断提高假新闻检测性能。这些研究结果表明，TED 非常适合在各种环境中部署，包括对特定模型的访问可能有限或首选开源模型的情况。

5.3 案例研究

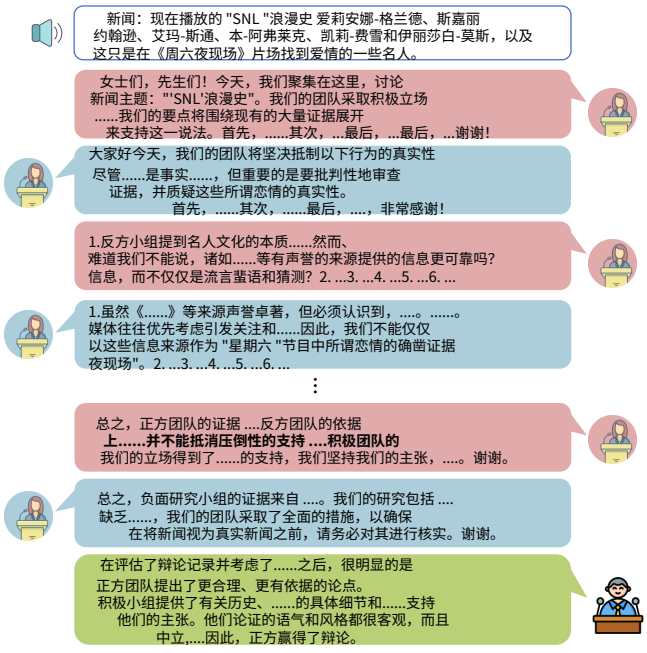


图 4：区域辩论案例研究及其辩论报告可作为可解释假新闻检测的依据。

图 4 展示了以假新闻为案例的辩论过程的微观比较分析。在这场辩论中，参赛者分为正方和反方两队，通过开篇、盘问和总结陈词完成辩论情景。辩论开始时，正方团队明确表示："本次辩论的首要目标是讨论新闻的真实性"，并提出关键点作为支持其立场的证据。反方也采取了类似的方法，提出了自己的论断。在提问和反驳阶段，双方都对对方的观点提出质疑，并对提出的问题做出回应。在总结陈词阶段，正方回顾并总结了之前的辩论，对反方的论点提出了质疑，并强化了自己的观点。反方虽然重申了自己的立场，但并没有对正方的观点提出深刻的质疑。最终，根据辩论的细节、语气和风格等因素，评委们决定支持正方。整个辩论过程形成了一个 "DebateFlow"，清晰地展示了评判过程。评委的辩论报告也可作为判断新闻是否为假新闻的依据，从而实现可解释的假新闻检测。这种可解释性对于现实生活中的编队验证至关重要，是 TED 框架的一大优势，也是其他模型难以实现的方面。

6 结论

总之，我们的研究介绍并评估了 TED 框架，这是一种新颖的假新闻检测方法，它将 LLM 的优势融入了结构化多代理辩论中。通过在 ARG-EN 和 ARG-CN 数据集上进行广泛的实验验证，TED 在多个性能指标上始终优于传统的机器学习和独立的 LLM。受人类辩论的启发，该框架的设计利用了双视角和结构化论证，为评估新闻条目的真实性提供了一种有效的机制。我们的工作不仅通过提高准确性和可解释性推动了假新闻检测领域的发展，还为开发负责任的信息系统提供了宝贵的见解。通过实现更可靠、更透明的内容评估，TED 有助于创建信息检索系统，更好地支持用户浏览和评估日益复杂的数字信息。

附录

在使用 ARG-CN 数据集的实验中，我们使用的提示是上述提示集的相应中文版本。所有其他函数都可以在我们的代码文件中找到，网址是：https://github.com/LiuYuHan31/TED_fake-news-detction。

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The Truth Becomes Clearer Through Debate! Multi-Agent Systems with Large Language Models Unmask Fake News

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Abstract

In today's digital environment, the rapid propagation of fake news via social networks poses significant social challenges. Most existing detection methods either employ traditional classification models, which suffer from low interpretability and limited generalization capabilities, or craft specific prompts for large language models (LLMs) to produce explanations and results directly, failing to leverage LLMs' reasoning abilities fully. Inspired by the saying that "truth becomes clearer through debate," our study introduces a novel multi-agent system with LLMs named TruEDebate (TED) to enhance the *interpretability* and *effectiveness* of fake news detection. TED employs a rigorous debate process inspired by formal debate settings. Central to our approach are two innovative components: the DebateFlow Agents and the InsightFlow Agents. The DebateFlow Agents organize agents into two teams, where one supports and the other challenges the truth of the news. These agents engage in opening statements, cross-examination, rebuttal, and closing statements, simulating a rigorous debate process akin to human discourse analysis, allowing for a thorough evaluation of news content. Concurrently, the InsightFlow Agents consist of two specialized sub-agents: the Synthesis Agent and the Analysis Agent. The Synthesis Agent summarizes the debates and provides an overarching viewpoint, ensuring a coherent and comprehensive evaluation. The Analysis Agent, which includes a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between debate roles and arguments using an attention mechanism, providing the final judgment. Our extensive experiments on two datasets, ARG-EN and ARG-CN, demonstrate that the TED framework surpasses traditional methods across various metrics and, more importantly, enhances interpretable fake news detection by illuminating logical reasoning and structured debate processes leading to accurate conclusions. We release our code to support Information systems that use structured debate within *responsible* information systems for improved decision-making[1].

CCS Concepts

• **Computing methodologies** → **Natural language processing**; • **Security and privacy** → **Human and societal aspects of security and privacy**.

Keywords

Fake News, Debate, Large Language Models, Multi-Agent System

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1 Introduction

The rapid propagation of fake news through social networks has become pervasive, significantly impacting individuals, vulnerable communities, and society. This widespread dissemination poses considerable challenges for the governance of public opinion [1], influencing elections [11], undermining public health policies [13], fairness [57–59] and threatening social stability [40]. Consequently, the urgency to develop effective and interpretable methods for detecting fake news has never been greater [37].

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[1] https://github.com/LiuYuHan31/TED_fake-news-detction

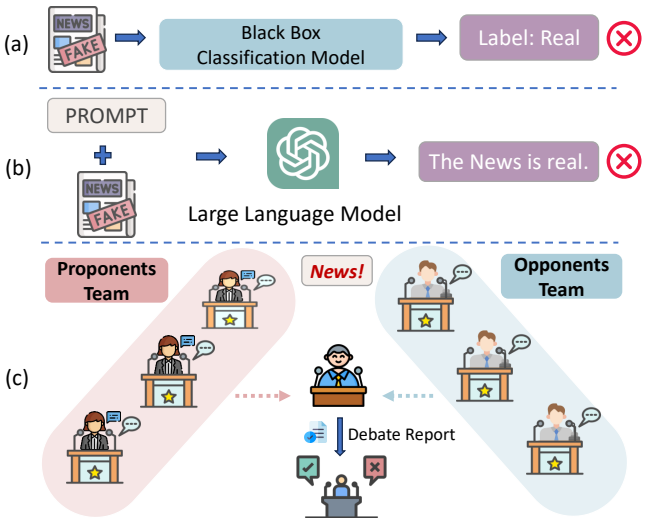


Figure 1: (a) Traditional models primarily rely on black box classification models for detecting fake news. (b) LLMs are prompted to output detection results directly, which do not fully utilize the potential of large model reasoning. (c) Our multi-agents framework simulates a formal debate scenario to engage in discussions on news topics, reaching detection results through debate reasoning and interpretability.

Traditional fake news detection methods predominantly rely on machine learning models, particularly those based on BERT [7], which excel in text analysis [16, 19, 31]. While these models have achieved notable success, they often suffer from low interpretability and limited generalization capabilities, hindering their effectiveness across diverse contexts. Recent approaches have attempted to leverage LLMs by developing specific prompts to generate explanations and detection results [15, 29]. However, these methods may not fully utilize LLMs' inherent reasoning abilities and complex cognitive processes, frequently resulting in excessively simplified and premature conclusions.

In the context of truth-seeking, the proverb "*truth becomes clearer through debate*" emphasizes the importance of examining and challenging information from multiple perspectives to arrive at a more accurate understanding [2, 21]. The exchange of differing viewpoints in a structured debate often leads to refining and clarifying ideas. However, existing fake news detection models fail to incorporate this approach fully, often limiting their analysis to a single, unchallenged perspective. This limitation prevents a comprehensive evaluation of news content and may overlook detailed signals that are critical for accurate detection.

To address these challenges, we introduce a novel multi-agent system with LLMs named TruEDebate (TED). TED is designed to enhance the interpretability and effectiveness of fake news detection by following the structured methodologies of formal debate. Inspired by the principle that truth emerges more clearly through debate, our approach simulates dynamic and structured debates among multiple agents, each examining the news from different angles. This debate structure follows the Lincoln-Douglas Debate format [8], where agents engage in a series of stages to present, challenge, and refine their arguments. This framework mirrors the human cognitive process of evaluating information through discourse, fostering deep intellectual engagement and comprehensive evaluation of news content. In this framework, multiple agents, each assigned distinct roles, engage in a structured debate process to evaluate the truth of news items. This debate process enhances the interpretation of the detection process and reflects the human cognitive process of debate, where conflicting viewpoints are analyzed and integrated.

The TED framework comprises two innovative components: the DebateFlow Agents and the InsightFlow Agents. The DebateFlow Agents organize multiple LLM-powered agents into two opposing teams—one supporting and the other opposing the truth of the news. These agents engage in sequential stages of opening, cross-examination rebuttal, and closing statements, effectively simulating a rigorous debate environment. This setup encourages the exploration of differing viewpoints, refining, and clarifying ideas, and addresses the limitation of existing models that analyze information from a single perspective. Concurrently, the InsightFlow Agents encompass two specialized sub-agents: the Synthesis Agent and the Analysis Agent. The Synthesis Agent is responsible for summarizing the debates and providing a comprehensive viewpoint, ensuring a coherent and thorough conclusion. The Analysis Agent, composed of a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between different debate roles and arguments through an interaction attention mechanism. By capturing the detailed relationships and discourse structures inherent in the debates, the Analysis Agent effectively identifies critical indicators essential for fake news detection.

We conducted comprehensive experiments on two datasets, ARG-EN and ARG-CN, to evaluate the effectiveness of TED. The results indicate that our framework outperforms traditional methods across multiple evaluation metrics. Additionally, TED facilitates structured debate procedures and enhances logical reasoning, contributing to accurate conclusions. By enabling systematic debate and in-depth analysis, our approach advances the development of information systems capable of making more precise and transparent decision-making processes, thereby supporting the creation of responsible information environments.

Our contribution can be summarized in the following ways:

- We introduce TED, a novel and interpretable fake news detection framework based on LLMs. TED not only performs detection through structured debates but also provides interpretable justifications to aid in understanding the results.
- Inspired by Lincoln-Douglas debate theory [8], which holds that "truth becomes clearer through debate," we design two key components within our framework: the Synthesis Agent and the Analysis Agent. The Synthesis Agent summarizes the debates and synthesizes comprehensive viewpoints, ensuring coherent conclusions. The Analysis Agent, incorporating a role-aware encoder and a debate graph, integrates role embeddings and models the interactions between different debate roles and arguments through an attention mechanism. This structure enables the Analysis Agent to identify critical indicators essential for accurate fake news detection.

• Extensive experiments on the ARG-EN and ARG-CN datasets demonstrate TED's superior performance to traditional methods, showing its ability to make accurate and transparent decisions. Furthermore, TED also maintains robust performance across different LLM backbones, showcasing its adaptability and generalizability in fake news detection.

2 Related Work

2.1 Fake News Detection

Fake news detection is an essential step in the fight against misinformation [12]. Earlier works in this area include the study by Qian et al. [36], which focuses on the early detection of fake news, considering only the text of news articles available at the time of detection. Yu et al. [49] applied a convolutional approach for misinformation identification. As the field evolved, various methods have been introduced to improve detection efficacy. For example, Jin et al. [17] move towards fine-grained reasoning for fake news detection by better reflecting the logical processes of human thinking and enabling the modeling of subtle clues. Furthermore, Zheng et al. [60] investigated the integration of diverse multi-modal data considering the complexity of their interrelations. Recently, with the rise of LLMs, Hu et al. [15] combined large and small LMs and achieved good improvements using only textual content for fake news detection. The agent workflow based on a large language model designed by Li et al. [25] uses tool learning and knowledge retrieval to detect fake news. Liu et al. [27, 28] employs a multi-agent system to simulate the propagation and evolution of fake news.

However, they did not fully utilize the potential of LLMs. Our research enhances the capability of fake news detection by further leveraging a debate system composed of multiple agents with LLMs. In our approach, we do not call any external tools, nor do we require knowledge-based retrieval, making it easily transferable to any fake news detection scenario.

2.2 Multi-agent Discussion with LLMs

Despite significant advancements in LLMs, research on multi-agent discussions, where multiple LLMs interactively collaborate to solve problems, is still in its early stages, aiming to leverage collective capabilities beyond individual models. A key contribution in this field is ChatEval [3], which designs a team of multi-agent referees to independently deliberate and assess the quality of responses generated by various models on open-ended questions and standard natural language generation tasks. Other studies emphasize diversifying agent roles to enrich the debate process [24, 33]. Liang et al. [26] further explores how role assignment impacts the effectiveness of multi-agent interactions. Chen et al. [4] introduce the ReConcile framework, which uses different LLMs and a weighted voting system to reach consensus, capturing a broader range of perspectives. Du et al. [9] employs multi-agent debate to improve the mathematical abilities of LLMs. Additionally, Zhang et al. [52] analyzes multi-agent discussions from a social psychology perspective to understand how social factors influence collaboration and reasoning processes among agents.

However, these works do not truly depict a debate process within their frameworks and cannot analyze problems from multiple perspectives. Building upon these foundations, our work introduces a structured debate framework that leverages multi-agent interactions to enhance the interpretability and effectiveness of fake news detection.

2.3 LLM-Based Multi-Agent System

With LLMs' powerful text generation capabilities, integrating LLMs into simulating social dynamics represents a burgeoning field of research, yielding promising results [18, 23, 39, 45]. These LLM-based generative agents excel in digital environments, demonstrating proficiency in natural language tasks [5, 6, 50] and reasoning task [51]. Törnberg et al. [42] used LLMs and agent-based modeling to simulate social media, focusing on news feed algorithms and offering real-world insights. Further, Park et al. [34] demonstrated that LLM-based agents could generate social media content indistinguishable from that produced by humans. Moreover, Qian et al. [35] designed ChatDev, a virtual development company that simulates a human development team.

These frameworks that utilize LLM multi-agent systems to solve complex tasks all share a crucial component: the interaction among multiple agents [3, 14, 30, 41, 46, 54–56]. However, these works consider only the message passing between different agents. While some claim to engage in multi-agent debates, they do not adequately depict the debate process during interactions. Our work establishes a standard debate scenario, where various roles debate fake news from different perspectives to determine the truth of the news topic. To the best of our knowledge, this is the first instance of a social debate simulation in the context of fake news.

3 Methodology

3.1 Task Formulation

We reformulate fake news detection as an interactive debate among multiple LLM agents. We introduce a system comprising N LLM agents denoted as $\mathcal{A} = (a_1, \ldots, a_N)$, each assigned unique roles and stances in the debate process. Given a news item F, all agents engage in a structured debate to determine its veracity. The agents are organized into two opposing teams:

- **Proponents**: Agents who argue that F is true.
- **Opponents**: Agents who argue that F is fake.

Each agent plays a specific role in the debate and participates in predefined stages. All interactions are recorded in a debate log D. At the conclusion stage of the debate, two types of agents evaluate the outcome. The Synthesis Agent analyzes the debate log D to produce a comprehensive summary report S. At the same time, the Analysis Agent utilizes the summary report S and the original news item F to predict the truth value.

Our objective is to leverage LLMs' reasoning capabilities within this debate framework to enhance the effectiveness and interpretability of fake news detection.

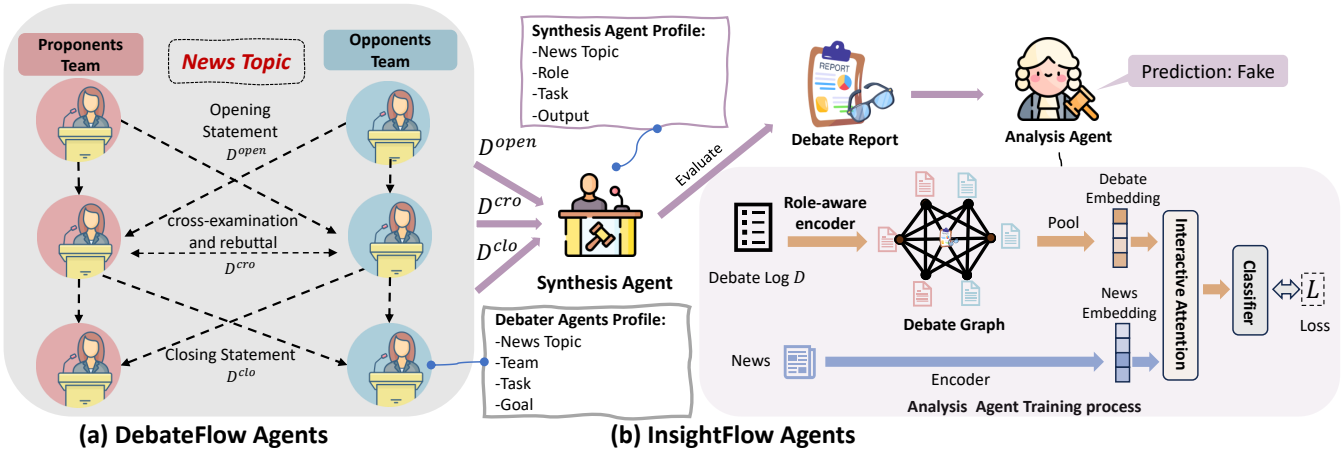


Figure 2: Our TED framework consists of two main components: (a) the DebateFlow Agents and (b) the InsightFlow Agents. The DebateFlow Agents simulate the reasoning process in a debate scenario, including the Opening, Cross-examination, Rebuttal, and Closing Statement. The InsightFlow Agents consist of the Synthesis Agent and the Analysis Agent. The former generates a debate report based on all debate records, which is then passed to the latter to predict whether the new content is true.

3.2 Our Multi-Agents Debate Framework

We introduce **TruEDebate (TED)**, a framework that enhances fake news detection by simulating a structured debate among multiple LLM-powered agents. TED consists of two main components:

• **DebateFlow Agents**: Organized into two opposing teams—the *Proponents*, who argue that the news item is true, and the *Opponents*, who argue that it is fake. Inspired by Lincoln-Douglas debate theory [8], these agents engage in a multi-stage debate process, including opening, cross-examination, rebuttal, and closing statements. This setup mirrors formal debate settings, encouraging thorough examination of the news from multiple perspectives.

• **InsightFlow Agents**: Comprising the *Synthesis Agent* and the *Analysis Agent*. The Synthesis Agent summarizes the debate, providing a cohesive overview of the key arguments and counterarguments. The Analysis Agent employs a role-aware encoder and a debate graph to model the interactions and discourse structures within the debate, ultimately predicting the authenticity of the news item.

As shown in Figure 2, through the interactions of these agents, TED fosters deep intellectual engagement and logical reasoning, leading to more accurate and interpretable fake news detection. The structured debate process not only refines the agents' understanding of the news item but also provides transparent justifications for the final decision, embodying the principle that "truth becomes clearer through debate."

3.3 DebateFlow Agents

The **DebateFlow Agents** form the core of the TED framework, orchestrating a structured debate to assess the authenticity of a news item F. The agents are divided into two opposing teams: the *Proponents* ($\mathcal{T}_{pro}$), who argue that F is true, and the *Opponents* ($\mathcal{T}_{opp}$), who contend that F is fake. Each team consists of agents assigned specific roles corresponding to different debate stages.

Inspired by Lincoln-Douglas debate [8], which is commonly used in debate settings to resolve controversial issues, the debate is structured into three main stages: *Opening*, *Cross-examination and Rebuttal*, and *Closing*. Let $\Phi = \{\phi_1, \phi_2, \phi_3\}$ denote these stages. In each stage $\phi_k \in \Phi$, agents from both teams participate according to their roles. In the *Opening* stage (ϕ_1), agents present their initial arguments supporting their team's stance. Each agent a_i generates an argument $d_i^{(1)}$ based on the news item F and their team's position:

$$d_i^{(1)} = f_{arg}(F, Stance_i), \tag{1}$$

where f_{arg} represents the argumentation prompt function implemented by the LLM, and $Stance_i \in \{True, Fake\}$ corresponds to the team's stance.

In the *Cross-examination and Rebuttal* stage (ϕ_2), agents challenge the opposing team's arguments through questions and rebuttals. An agent a_j from one team formulates a question or rebuttal $d_j^{(2)}$ targeting the arguments made in the previous stage by the opposing team:

$$d_j^{(2)} = f_{reb}(D^{(1)}, Stance_j), \tag{2}$$

where f_{reb} is the Cross-examination and Rebuttal prompt function, and $D^{(1)}$ is the set of arguments in stage ϕ_1.

In the *Closing* stage (ϕ_3), agents summarize their team's position and reinforce key points. Each agent a_k produces a concluding statement $d_k^{(3)}$:

$$d_k^{(3)} = f_{clo}(D^{(1)} \cup D^{(2)}, Stance_k), \tag{3}$$

where f_{clo} is the closing prompt function, and $D^{(1)} \cup D^{(2)}$ represents the accumulated debate content from previous stages.

The debate log D accumulates all interactions during the debate:

$$D = \bigcup_{k=1}^{3} D^{(k)}, \quad \text{with} \quad D^{(k)} = \{d_i^{(k)} \mid a_i \in \mathcal{A}^{(k)}\}, \tag{4}$$

Input: News item F
Output: Predicted truth value $\hat{y}$ of news item F, Explanatory reasons R
Initialize DebateFlow Agents:
Assign agents into two teams:
- *Proponents*: Support that F is true
- *Opponents*: Argue that F is fake

Assign roles within each team (e.g., Opening Speaker, Questioner, Closing Speaker)
DebateFlow Agents:
for each debate stage ϕ in [Opening Statement, Cross-examination, Rebuttal, and Closing Statement] **do**
  Agent i presents arguments or rebuttals based on the role
  Record interactions in debate log D_ϕ
end for
InsightFlow Agents:
Synthesis Agent:
Analyze debate logs $D = \{D_\phi\}$ to produce evaluation report S
Analysis Agent:
for each debate interaction (node) n_i in D **do**
  Obtain node n_i embedding h_i with a Role-aware Encoder
end for
Build Dabate graph $G = (V, E)$ with nodes $V = \{h_i\}$
Apply GATs to G to obtain graph embedding g
Encode news item F to obtain content embedding e_F
Combined representation $c \leftarrow \text{InteractiveAttention}(g, e_F)$
Predict truth value $\hat{y}$ using c
return Predicted truth value $\hat{y}$, explanatory reasons $R = \{S, D\}$

Algorithm 1: TruEDebate (TED): Multi-Agent Debate Framework for Fake News Detection

where $\mathcal{A}^{(k)}$ is the set of agents participating in stage ϕ_k.

Each agent's contribution $d_i^{(k)}$ depends on the news item F, their team's stance, their role in the debate, and the prior debate context $D^{<k}$:

$$d_i^{(k)} = f_{\text{LLM}}(F, D^{<k}, \text{Role}_i, \text{Stance}_i), \tag{5}$$

where $D^{<k} = \bigcup_{m=1}^{k-1} D^{(m)}$ represents the debate history up to stage ϕ_{k-1}, and Role_i specifies the agent's role.

The entire algorithm process can be described in Algorithm 1. This structured interaction ensures that agents consider both the news content and the evolving debate context, promoting coherence and depth in their reasoning. By engaging in this process, the DebateFlow Agents simulate a rigorous examination of the news item F, mirroring the dynamics of human debates and fostering a comprehensive evaluation from multiple perspectives.

The whole set of function prompts can be found in our code files.

3.4 InsightFlow Agents

After the debate conducted by DebateFlow Agents, the **InsightFlow Agents** analyze the accumulated discourse to produce a final prediction on the authenticity of the news item F. This component consists of two principal agents: the **Synthesis Agent** and the **Analysis Agent**.

3.4.1 ***Synthesis Agent.*** The Synthesis Agent processes the debate log D, which contains all interactions from the debate stages, to generate a concise and comprehensive summary S of the key arguments and counterarguments presented by both teams. This summary is evaluated from multiple perspectives of fake news detection to ensure a well-rounded judgment. Formally, the core function prompt of the Synthesis Agent is defined as follows:

> It should contain a detailed explanation of your assessment of the debate. Focus on evaluating the authenticity of the news involved in the topic by checking the following:
> 1. Whether the news contains specific details and verifiable information.
> 2. Whether the news cites reliable sources or news organizations.
> 3. The tone and style of the news, with real news generally being more objective and neutral.
> 4. Any use of emotional language, which might be a characteristic of fake news.
> 5. Whether the information in the news can be confirmed through other reliable channels.

The purpose of the Synthesis Agent is to encapsulate the critical points of the debate, providing a coherent overview highlighting the strengths and weaknesses of each side's arguments. This summary S bridges the debate discourse and the subsequent analytical processes, ensuring that the most pertinent information is readily accessible for further evaluation.

3.4.2 ***Analysis Agent.*** The Analysis Agent integrates the debate interactions with the original news content to predict the truth value $\hat{y}$ of the news item F. This agent employs a combination of role-aware encoder, debate graph, and a news-debate interactive attention mechanism to capture both semantic content and the relational dynamics of the debate, as well as their relevance to the news content.

Role-aware Encoder. To capture the distinct characteristics of different debate roles, we design a role-aware encoder. For each debate interaction d_i, the node representation is constructed by concatenating the encoder's `[CLS]` token embedding with a projected role embedding:

$$\begin{aligned} \mathbf{h}_i^{\text{enc}} &= \text{Encoder}(\text{Text}_i), \\ \mathbf{r}_i^{\text{proj}} &= \mathbf{W}_{\text{role}}\, \mathbf{e}_i, \\ \mathbf{h}_i^{\text{node}} &= \left[\, \mathbf{h}_i^{\text{enc}} \,;\, \mathbf{r}_i^{\text{proj}} \,\right], \end{aligned} \tag{6}$$

where $\mathbf{h}_i^{\text{enc}} \in \mathbb{R}^{d_h}$ is the `[CLS]` token representation from the encoder, $\mathbf{e}_i \in \mathbb{R}^{d_r}$ is the embedding for role r_i, $\mathbf{W}_{\text{role}} \in \mathbb{R}^{d_h \times d_r}$ projects the role embedding to match the encoder's dimension, and $[\,;\,]$ denotes vector concatenation. The resulting node representation $\mathbf{h}_i^{\text{node}}$ integrates both semantic content and role-specific information of each debate interaction.

Debate Graph. The encoded node representations are structured into a debate graph $G = (V, E)$, where the nodes $V = \{h_i^{\text{node}}\}$

represent the debate interactions, and the edges E represent the relationships between interactions, such as sequential order or explicit references.

We employ Graph Attention Networks (GAT) [43] to process the node representations. The GAT layers update each node's representation by attending to its neighbors:

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l)}\mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\right), \tag{7}$$

where $\mathbf{h}_i^{(l)}$ is the representation of node i at layer l, $\mathcal{N}(i)$ denotes the set of neighbors of node i, $\mathbf{W}^{(l)}$ is a learnable weight matrix, σ is an activation function, and $\alpha_{ij}^{(l)}$ are the attention coefficients. By stacking multiple GAT layers, the model captures higher-order interactions and the relational dynamics within the debate.

Debate-News Interactive Attention. To integrate the debate representations with the news content, we employ an interactive attention mechanism. First, we encode the news content F and apply global pooling to obtain the graph-level debate representation $\mathbf{g}$:

$$\begin{aligned} \mathbf{e}_F &= \text{Encoder}(F), \\ \mathbf{g} &= \text{GlobalPool}(\mathbf{H}^{(L)}), \end{aligned} \tag{8}$$

where $\mathbf{H}^{(L)}$ is the output of the L-th GAT layer, and `GlobalPool` denotes a global mean pooling operation.

Next, we project both the debate representation and the news content embedding to a common dimensionality:

$$\begin{aligned} \mathbf{g}^{\text{proj}} &= \mathbf{W}_g\mathbf{g}, \\ \mathbf{e}_F^{\text{proj}} &= \mathbf{W}_e\mathbf{e}_F, \end{aligned} \tag{9}$$

where $\mathbf{W}_g$ and $\mathbf{W}_e$ are projection matrices.

We then compute the interaction representation $\mathbf{c}$ using multi-head attention (MHA) to capture the interactions between the debate and news content:

$$\mathbf{c} = \text{MHA}(\mathbf{e}_F^{\text{proj}}, \mathbf{g}^{\text{proj}}, \mathbf{g}^{\text{proj}}), \tag{10}$$

Finally, we concatenate the projected debate representation with the interaction representation to obtain the comprehensive representation:

$$\mathbf{h} = [\mathbf{g}^{\text{proj}}; \mathbf{c}]. \tag{11}$$

This interactive attention allows the model to focus on different aspects of the debate relevant to the news content, effectively integrating information from both sources.

Agent Training. The combined representation $\mathbf{h}$ is processed through a fully connected layer followed by a softmax activation to generate the predicted probability distribution over classes. The model is trained by minimizing the cross-entropy loss between the predicted probabilities and the actual labels:

$$\begin{aligned} \hat{\mathbf{y}} &= \text{softmax}(\mathbf{W}_{\text{fc}}\mathbf{h} + \mathbf{b}_{\text{fc}}), \\ \mathcal{L} &= -\sum_k y_k \log \hat{y}_k, \end{aligned} \tag{12}$$

where $\mathbf{W}_{\text{fc}}$ and $\mathbf{b}_{\text{fc}}$ denote the weights and bias of the classification layer, $\hat{\mathbf{y}}$ represents the predicted probability distribution, y_k is the ground truth label for class k, and $\hat{y}_k$ is the predicted probability for class k.

4 Experiment

4.1 Experiment Setup

4.1.1 Datasets. For evaluation, we utilized two datasets: ARG-EN and ARG-CN [15]. These datasets are derived from the Chinese dataset Weibo21 [32] and the English dataset GossipCop [38]. In previous work [15], they processed these datasets by deduplication and dividing the data temporally. This approach helps prevent overestimating the SLM performance due to potential data leakage. We used these two datasets in the experiment. The statistical details of the datasets are summarized in Table 2.

4.1.2 Baselines. Following previous work [15], we conducted experiments comparing three types of methods.

• **LLM-Only Methods**: Directly use GPT-3.5-turbo and GPT-4o-mini as a LLM for fake news detection.

• **SLM-Only Methods**: Small Language Models (SLMs), like BERT-based models, generally perform well on fake news detection tasks. In this category, we include the following approaches:

(1) Bert [7]: Detecting fake news using a fine-tuned vanilla BERT-base model.

(2) EANN [44]: This model employs auxiliary adversarial training to effectively isolate and reduce event-related features, using the publication year as the label for the auxiliary task.

(3) Publisher-Emo [53]: A model that integrates a range of emotional attributes with textual features to detect fake news.

(4) ENDEFA [61]: A model that employs causal learning to eliminate entity bias, enhancing its generalization capabilities on fake news datasets with distribution shifts. All approaches in this category utilize the same SLM (BERT) for text encoding.

• **LLM+SLM Methods**. LLM solely often does not perform well on the task. Hence, a number of works combine LLM with SLM to enhance performance.

(1) Bert+Rationale: It combines features from both the news and the rationale encoder, feeding them into an MLP for prediction.

(2) SuperICL [47]: It utilizes the SLM as a plug-in to enhance the in-context learning capabilities of the LLM by incorporating predictions and confidence levels for each test sample into the prompt.

(3) ARG [15]: This model designs an Adaptive Rationale Guidance network for fake news detection, where SLMs selectively derive insights on news analysis from the rationales provided by LLMs.

(4) ARG-D [15]: It is a rationale-free version of ARG created through distillation, designed for cost-sensitive scenarios that do not require querying LLMs.

4.1.3 Implementation Details. We use Python to operationalize our framework TED. The LLM used is `GPT-4o-mini` accessed via OpenAI API calls due to its representativeness and convenient calling. Agents and the scene they live in were defined using a Python library Mesa [20]. For Analysis Agent, We utilize a fine-tuned BERT model due to its extensive application in this domain as highlighted by recent studies [19]. For our experiments, we employ models from the Transformers package *chinese-bert-wwm-ext* for Chinese and *bert-base-uncased* for English evaluations. Optimization is performed using Adam [22], with a comprehensive grid search to

Table 1: The performance of the TED, as well as the LLM-only, SLM-only, and LLM+SLM methods was assessed. All results (except for those of `GPT-4o-mini` and ChatEval) for the baselines are taken from Hu et al. [15]. The top two results for both macF1-score and accuracy are highlighted, with the highest in each category being bolded and the second highest underlined. Numbers in bold mean that the improvements to the baseline models are statistically significant (t-test with p-value<0.01).

Model		ARG-EN				ARG-CN			
		macF1	Acc.	$F1_{\text{real}}$	$F1_{\text{fake}}$	macF1	Acc.	$F1_{\text{real}}$	$F1_{\text{fake}}$
LLM-Only	GPT-3.5-turbo	0.702	0.813	0.884	0.519	0.725	0.734	0.774	0.676
	GPT-4o-mini	0.691	0.845	0.909	0.472	0.725	0.746	0.780	0.670
SLM-Only	Bert	0.765	0.862	0.916	0.615	0.753	0.754	0.769	0.737
	EANN	0.763	0.864	0.918	0.608	0.754	0.756	0.773	0.736
	Publisher-Emo	0.766	0.868	0.920	0.611	0.761	0.763	0.784	0.738
	ENDEF	0.768	0.865	0.918	0.618	0.765	0.766	0.779	0.751
LLM+SLM	Bert + Rationale	0.777	0.870	0.921	0.633	0.767	0.769	0.787	0.748
	SuperICL	0.736	0.864	0.920	0.551	0.757	0.759	0.779	0.734
	ARG	0.790	0.878	0.926	0.653	0.784	0.786	0.804	0.764
	ARG-D	0.778	0.870	0.921	0.634	0.771	0.772	0.785	0.756
Multi-Agents	ChatEval(One-by-One)	0.733	0.860	0.919	0.546	0.694	0.717	0.778	0.611
	ChatEval(Simultaneous-Talk)	0.738	0.869	0.923	0.553	0.694	0.719	0.780	0.608
	TED(Ours)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**

Table 2: Statistics of the fake news detection datasets.

	ARG-CN			ARG-EN		
	Train	**Val**	**Test**	**Train**	**Val**	**Test**
Real	2,331	1,172	1,137	2,878	1,030	1,024
Fake	2,873	779	814	1,006	244	234
Total	5,204	1,951	1,951	3,884	1,274	1,258

determine the optimal learning rate. Performance is evaluated using the best-validation checkpoint from the testing results. For the Analysis Agent, the experiments were conducted using PyTorch, with the HuggingFace Transformers library for pre-trained models and PyTorch Geometric for graph neural network components. Text data was tokenized using the BERT tokenizer appropriate for the dataset's language. Debate interactions were encoded with role-aware embeddings, and news content was separately encoded using the BERT model. Additionally, a fixed graph structure was employed for each debate instance, with nodes representing debate interactions and edges capturing the logical flow and relationships between these interactions. Edges were predefined based on the debate sequence and interactions between roles. All experiments were run on machines equipped with NVIDIA 3090 GPUs to accelerate training. The specific experimental implementation details can be found in our code.

4.2 Main Results

Table 1 presents the performance of our proposed TED framework and various other models, such as LLM-only, SLM-only, and LLM+SLM methods, on the ARG-EN and ARG-CN datasets. From the results, we observe: (1) TED outperforms all other methods, achieving the highest macF1 scores and accuracy across the datasets, showcasing its exceptional capability in detecting fake news. This underlines the efficacy of our multi-agent debate method in thoroughly analyzing and judging the truth of news items. (2) The LLM-only method, which utilizes LLMs without structured argumentation, exhibits inferior performance compared to TED. This emphasizes the limited capability of LLMs when deployed in isolation, lacking the structured interaction and discourse mechanisms provided in TED. (3) Comparisons among SLM-only methods reveal incremental improvements across each variant, yet none achieve the performance level of TED. This suggests that while traditional SLMs are capable, they fall short of the robust and comprehensive analytical framework offered by the multi-agent system in TED. (4) The LLM+SLM methods, attempting to combine the capabilities of LLMs with SLMs, show varied performance. For instance, the ARG model demonstrates considerable effectiveness but still does not reach the level of TED. This indicates that while the integration of LLM and SLM provides benefits, the structured debate format of TED is crucial for optimal performance.

In addition, to evaluate the effectiveness of our TED framework for fake news detection, we compared it with the ChatEval framework [3], which uses *One-by-One* and *Simultaneous-Talk* strategies for multi-agent debates. Applying these strategies to our fake news detection task, as shown in Table 1, TED consistently outperforms ChatEval across all metrics. The formal debate structure of TED promotes more interactive reasoning among agents, allowing for a deeper analysis of news content. In contrast, ChatEval's strategies do not achieve the same depth in fake news detection. These results highlight TED's superiority in enhancing the accuracy and reliability of multi-agent systems for fake news detection. By fostering structured and dynamic interactions, TED not only improves detection accuracy but also ensures a more thorough examination of the news item.

Table 3: Ablation study of our model TED on ARG-EN and ARG-CN datasets. Numbers in bold mean that the improvements to the ablation models are statistically significant (t-test with p-value<0.01).

	ARG-EN				ARG-CN			
	macF1	Acc.	$F1_{\text{real}}$	$F1_{\text{fake}}$	macF1	Acc.	$F1_{\text{real}}$	$F1_{\text{fake}}$
Full Model (TED)	**0.803**	**0.892**	**0.932**	**0.674**	**0.795**	**0.798**	**0.815**	**0.774**
w/o DebateFlow Agents	0.780	0.874	0.918	0.642	0.767	0.781	0.795	0.739
w/o Synthesis Agent	0.792	0.881	0.924	0.660	0.771	0.787	0.801	0.741
w/o Analysis Agent	0.729	0.856	0.915	0.543	0.737	0.736	0.785	0.689

5 Analysis and Discussion

5.1 Ablation Study

We conducted an ablation study to assess the impact of each component in the TED framework. The results indicate that removing the DebateFlow Agents reduces macF1 scores by approximately 2.3% on ARG-EN and 2.8% on ARG-CN. This decline underscores the importance of the structured debate process in facilitating comprehensive multi-perspective analysis, which is essential for accurately determining the authenticity of news items. Removing the Synthesis Agent results in a moderate decrease in macF1 scores, dropping by 1.1% on ARG-EN and 2.4% on ARG-CN. This highlights the critical role of the Synthesis Agent in consolidating debate interactions into coherent summaries, thereby enhancing the model's ability to integrate diverse arguments effectively for precise classification. Omitting the Analysis Agent causes the most significant drop, with macF1 scores falling by 8.0% on ARG-EN and 4.5% on ARG-CN, emphasizing its crucial function in integrating debate insights with the news content. Overall, the ablation results confirm that each component of the TED framework is indispensable for achieving high performance in fake news detection. The DebateFlow Agents enable robust multi-agent interactions; the Synthesis Agent ensures effective consolidation of debate outcomes, and the Analysis Agent integrates these insights with the news item to get precise results.

5.2 Performance on Different Backbones

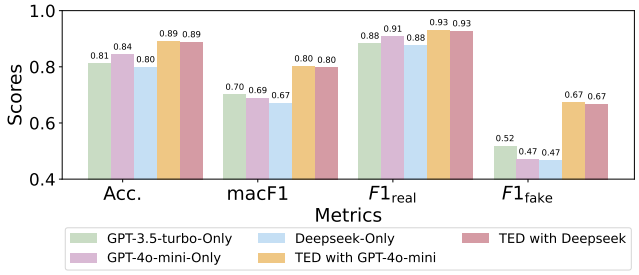


Figure 3: Performance of our framework TED with GPT-4o-mini and Deepseek as closed-source backbones respectively.

5.2.1 Performance on closed-source models. To evaluate the robustness and versatility of the TED framework, we conducted experiments on the ARG-EN dataset by replacing the DebateFlow Agents and Synthesis Agent with different closed-source models, namely GPT-4o-mini and Deepseek[2]. Figure 3 demonstrates that TED consistently enhances performance compared to the baseline LLM-Only model across all evaluation metrics, regardless of the specific closed-source model used. This outcome highlights the framework's adaptability to different large-scale models, showing that TED effectively leverages the unique strengths of various closed-source backbones. Additionally, we observed that the performance of TED with GPT-4o-mini and TED with Deepseek was comparable, with a slight advantage in favor of GPT-4o-mini. This suggests that while the two models are competitive, GPT-4o-mini appears to have a slight edge in some aspects, particularly in handling complex reasoning tasks. Despite this, the results across both backbones emphasize TED's robustness in boosting fake news detection performance regardless of the specific closed-source model used. These findings underline that TED is not limited to a single backbone and can effectively enhance performance across different closed-source models. The slight superiority of TED with GPT-4o-mini reinforces the notion that TED can leverage the inherent strengths of each model's reasoning abilities. This adaptability ensures that TED remains effective and reliable in real-world applications.

Overall, the TED framework proves to be both effective and adaptable, maintaining strong performance across a variety of closed-source models. These results validate TED's utility in diverse application scenarios, making it a highly versatile tool for fake news detection.

Table 4: Performance of TED with Qwen2.5 and Llama3.1 family models as backbones, respectively. Numbers with underlined mean the best performance with an open-source model and are statistically significant(t-test with p-value<0.01).

Backbones	**#Param**	**macF1**	**Acc**	$F1_{\textbf{real}}$	$F1_{\textbf{fake}}$
GPT-4o-mini	-	0.803	0.892	0.932	0.674
Qwen 2.5 [48]	7B	**0.794**	**0.887**	**0.926**	**0.661**
Llama 3.1 [10]	8B	0.787	0.879	0.920	0.653
Llama 3.1 [10]	70B	0.790	0.884	0.923	0.657

5.2.2 Performance on open-source models. To further evaluate the adaptability and robustness of the TED framework, we tested its performance on the ARG-EN dataset using several open-source language models, including Qwen-2.5 [48] and the Llama 3.1 family [10]. As demonstrated in Table 4, TED consistently improves

[2]https://www.deepseek.com/

performance across all the open-source models, highlighting its ability to significantly enhance fake news detection regardless of the specific language model used. This shows that TED is not constrained by the availability or type of backbone model, making it a versatile and effective tool for various real-world applications. Additionally, TED's performance with Qwen 2.5 slightly outperforms the Llama 3.1 variants. Although the differences in performance are minimal, this suggests that Qwen 2.5 is particularly well-suited to TED's debate-driven framework. It may offer a better handling of complex argumentation and context, which is essential for the structured evaluation TED relies on. On the other hand, TED with the Llama 3.1 models also performs admirably, and all configurations lead to better results than the baseline methods, further validating TED's effectiveness across different models. Although the performance with open-source models is lower than the results obtained with GPT-4o-mini, this can likely be attributed to differences in the underlying reasoning capabilities of the models. Open-source models, while highly capable, may not exhibit the same level of nuanced reasoning as closed-source models like GPT-4o-mini. Nevertheless, TED continues to produce strong results even with these open-source backbones, demonstrating that it can still leverage each model's strengths to achieve high levels of accuracy and reliable detection.

Overall, the results highlight TED's robustness and versatility. It consistently enhances fake news detection performance. These findings suggest that TED is well-suited for deployment in various settings, including scenarios where access to specific models may be limited or open-source models are preferred.

5.3 Case Study

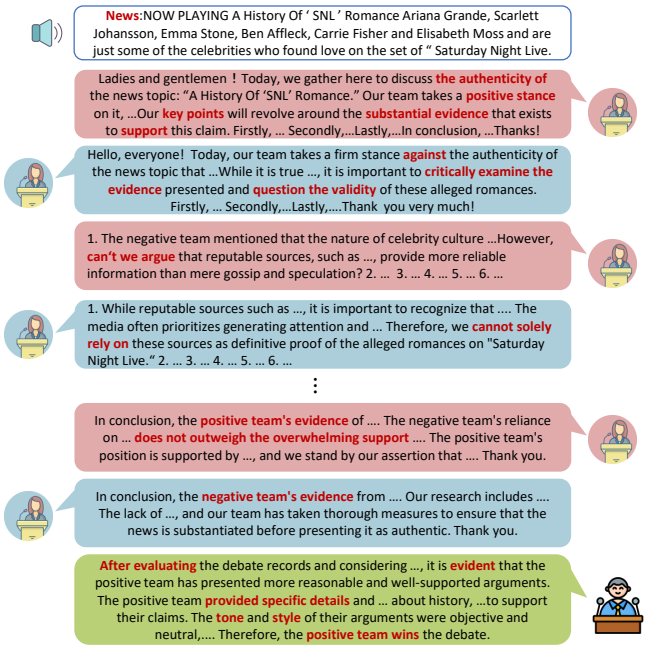


Figure 4: A real debate case study, with its debate report, can serve as justification for interpretable fake news detection.

Figure 4 presents a comparative micro-analysis of a debate process on fake news as a case study. Participants are divided into pro and con teams in this debate, completing the debate scenario through opening, cross-examinations, and closing statements. At the beginning of the debate, the pro team clearly stated, "The primary objective of this debate is to discuss the truth of the news," and they presented key points as evidence to support their stance. The con team adopted a similar approach, presenting their assertions. During the Questioning and Rebuttal phase, both sides challenged each other's viewpoints and responded to the queries raised. In the Closing Statement phase, the pro team reviewed and summarized the prior debate questioned the con team's arguments and reinforced their own points. Although the con team reiterated their position, they did not profoundly challenge the pro team's viewpoints. Ultimately, based on factors such as the debate's detail, tone, and style, the judges decided to favor the pro team. This entire debate process forms a "DebateFlow," clearly demonstrating the reasoning process. The judges' debate report also serves as a basis for determining whether news is fake, thus achieving an interpretable fake news detection. Such interpretability is crucial for real-life information verification, representing a significant advantage of the TED framework, an aspect that other models struggle to achieve.

6 Conclusion

In conclusion, our study introduces and evaluates the TED framework, a novel approach to fake news detection that incorporates the strengths of LLMs within a structured multi-agent debate. Through extensive experimental validation on the ARG-EN and ARG-CN datasets, TED consistently outperformed traditional machine learning and standalone LLMs across multiple performance metrics. Inspired by human debate, the framework's design leverages diverse perspectives and structured argumentation, providing a robust mechanism for assessing the truth of news items. Our work not only advances the field of fake news detection by enhancing accuracy and interpretability but also offers valuable insights for developing responsible information systems. By enabling more reliable and transparent content evaluation, TED contributes to creating information retrieval systems that can better support users in navigating and assessing the growing complexity of digital information.

A Appendix

In the experiment using the ARG-CN dataset, the prompt we used was the corresponding Chinese version of the aforementioned prompt set. And all other function prompts can be found in our code files at https://github.com/LiuYuHan31/TED_fake-news-detction.

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